



Tiller to aim gun

▼ SWEDISH SWIVEL GUN

Date c.1500

Origin Sweden

Material Iron

Shot Round or grapeshot

This is an example of a naval swivel gun. A lead ball would have been placed in the breech of the barrel and a powder chamber, usually made of iron and shaped like a tankard, placed in the trough behind it.

Trunnions, rounded projections on the sides of the barrel, help to elevate and lower muzzle

Channel for breech chamber



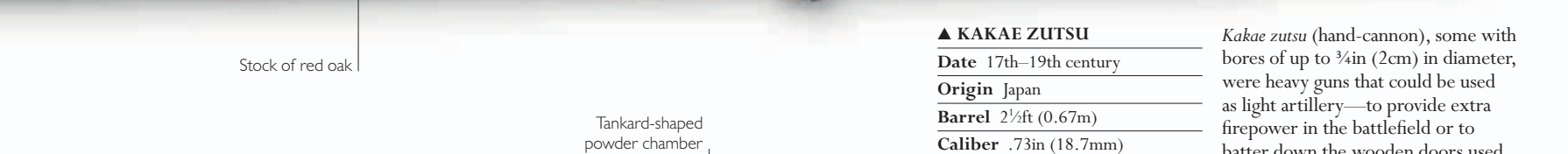
Joint of barrel and breech

Decorative expanded muzzle

Barrel decorated with a dragon in silver



Stock of red oak



Tankard-shaped powder chamber

▲ KAKAE ZUTSU

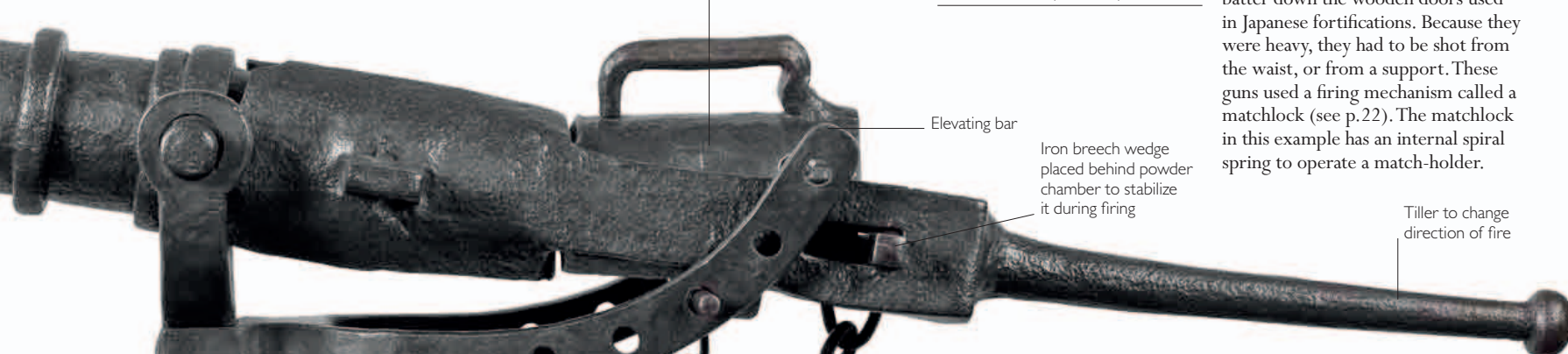
Date 17th–19th century

Origin Japan

Barrel 2½ft (0.67m)

Caliber .73in (18.7mm)

Kakae zutsu (hand-cannon), some with bores of up to ¾in (2cm) in diameter, were heavy guns that could be used as light artillery—to provide extra firepower in the battlefield or to batter down the wooden doors used in Japanese fortifications. Because they were heavy, they had to be shot from the waist, or from a support. These guns used a firing mechanism called a matchlock (see p.22). The matchlock in this example has an internal spiral spring to operate a match-holder.



Elevating bar

Iron breech wedge placed behind powder chamber to stabilize it during firing

Tiller to change direction of fire



Mounting podium

Chain secures breech wedge in place

▲ ENGLISH SWIVEL GUN

Date Late 15th century

Origin England

Length 4½ft (1.36m)

Caliber 2in (51mm)

Swivel guns were frequently employed for naval use. This model was mounted on the gunwales (upper edges of the sides) of a ship, where the superior arc of fire could be used to rake enemy vessels. Like most swivel guns, it is a breech-loader. As pictured, the breech chamber was wedged to hold it in place at the time of firing. This was true of almost all breech-loaders until the end of the 17th century.